THE UNIVERSITY OF MELBOURNE

COMP30023 – Computer Systems

Subject Overview

Maria Rodriguez Read

Semester 1, 2025

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Teaching Team

Maria Rodriguez Read – Lecturer (weeks 1 – 6)

- Consultation hours
  - Tue 11:15am - 12:15pm, MC level 5 Room 5211
  - Thu 11:15am - 12:15pm, MC level 5 Room 5210
  - Zoom available on demand

Lachlan Andrew – Lecturer (weeks 6 – 12)

- Consultation hours
  - Thu 1pm - 2pm, MC and Zoom
  - Fri 1pm - 2pm, MC and Zoom

Qifan Deng – Head tutor

11 Tutors (Check *Meet the Staff* page on Canvas)

What is a Computer System?

- What hardware and software components are involved when you access, view, and create content on social media?

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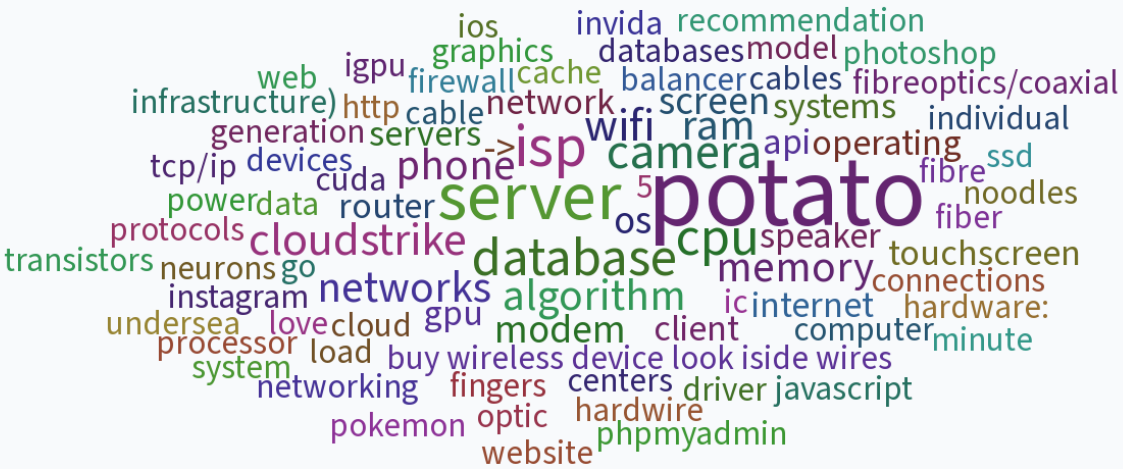
What is a Computer System?

Hardware and Software components involved when using social media



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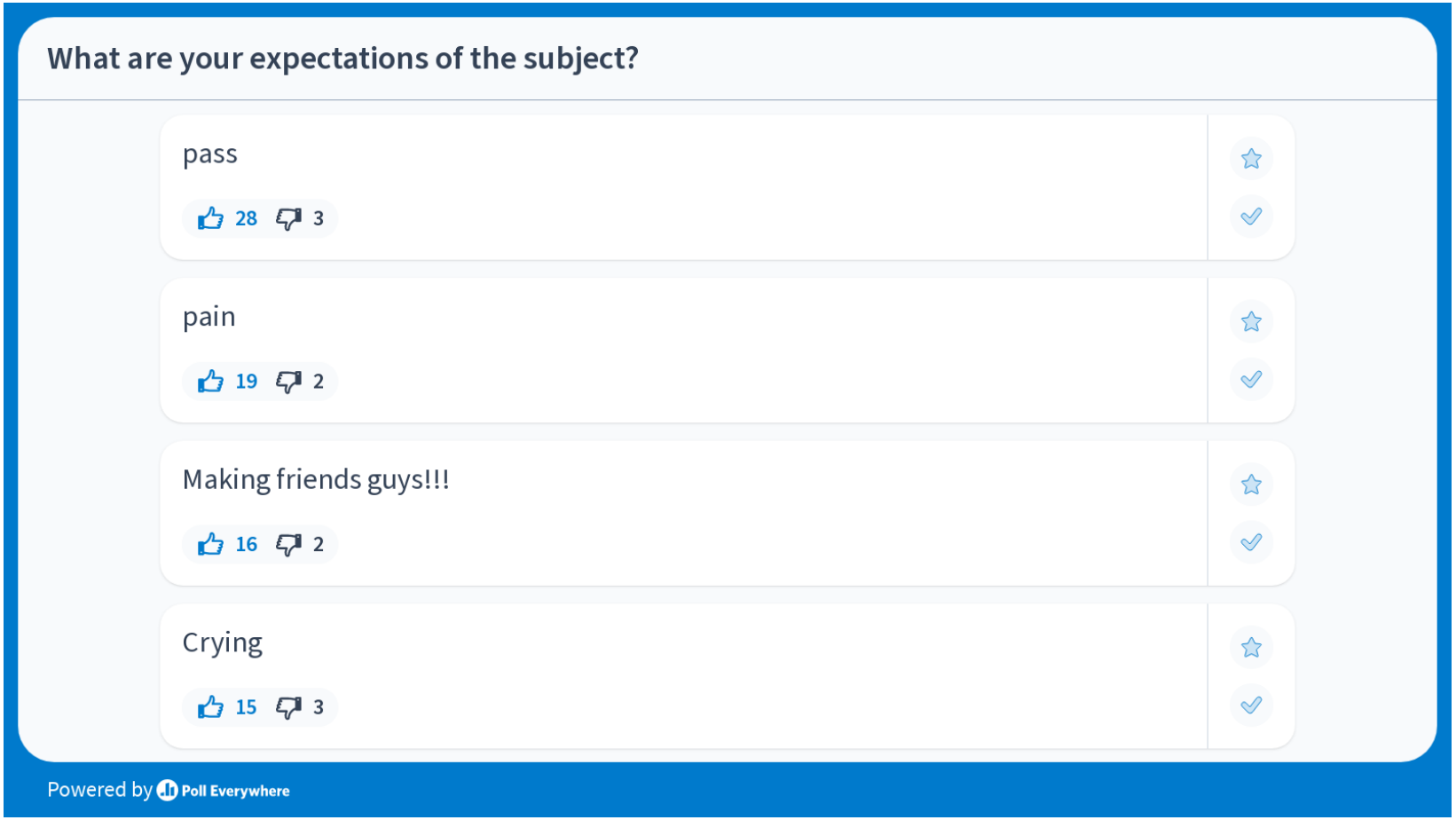
Your Expectations?



Intended Learning Outcomes

- ILO 1 - Demonstrate their knowledge of **operating systems** from the programmer's perspective
- ILO 2 - Demonstrate their knowledge of **networking technologies** from the programmer's perspective
- ILO 3 - Apply this knowledge to select appropriate tools and technologies for a problem at hand
- ILO 4 - Build simple server applications
- ILO 5 - Recognise some common **security** vulnerabilities and the reason why they are present in a particular system

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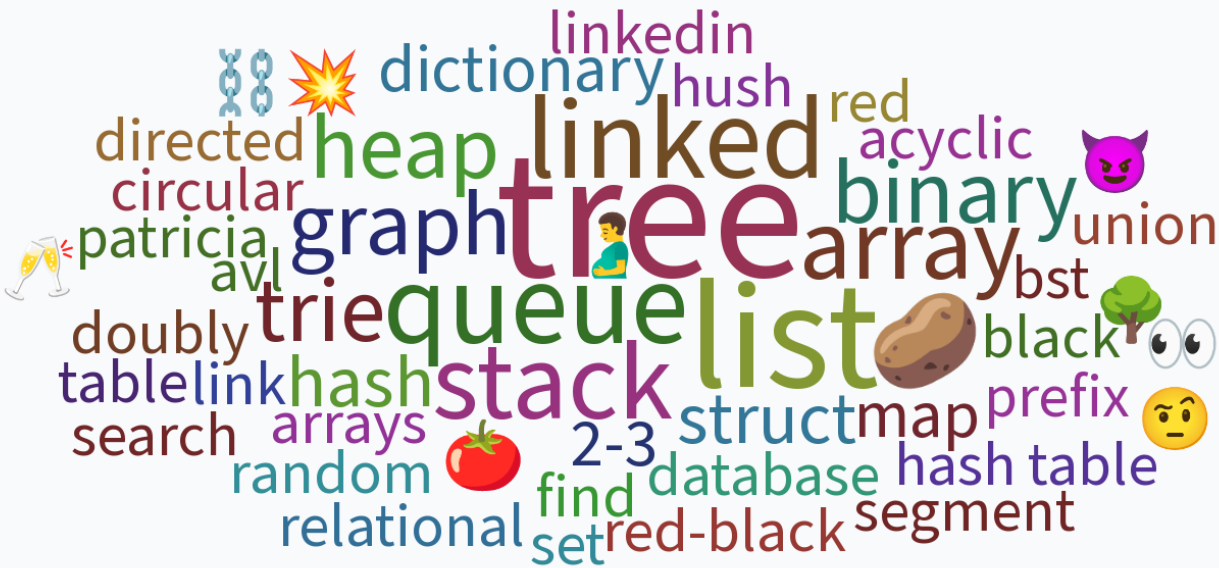
Data Structures?

Name a data structure commonly used in CS



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Prerequisites

One of:

- COMP20003 – Algorithms and Data Structures
- COMP20007 – Design of Algorithms
- COMP90038 – Algorithms and Complexity

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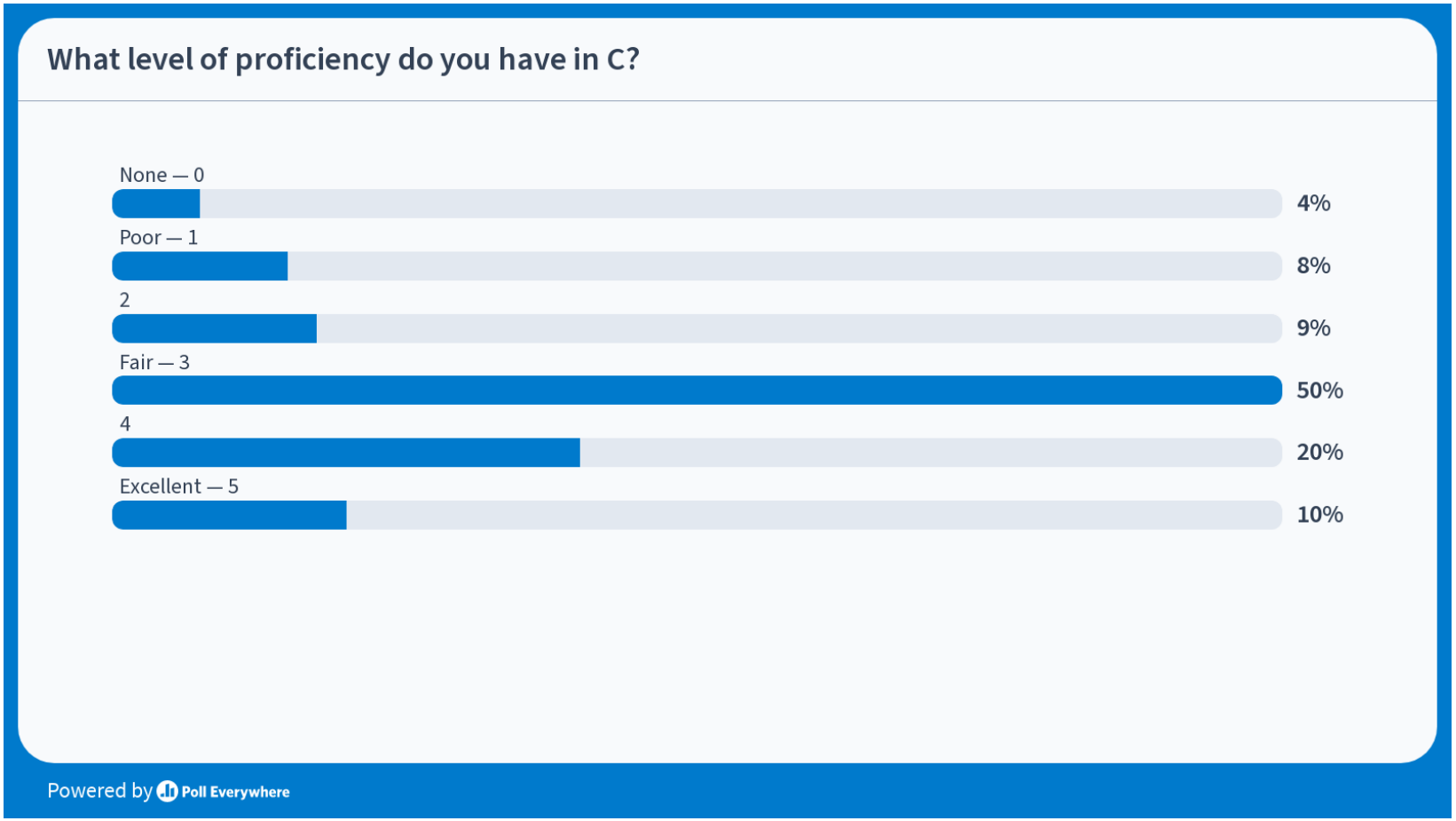
How familiar are you with C?



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Delivery

Lectures – F2F

- Tuesday 10am – 11am
- Thursday 2pm – 3pm
- Arts West (148B), Kathleen Fitzpatrick Theatre (101B)

Workshops – F2F

- 1st hour: Tutorial
- 2nd hour: Practical
- Attendance strongly encouraged – access code to weekly quiz

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Tentative Schedule

Week	Topic	Weekly Material	Notes	Projects	Quiz due Date
1 3 Mar - 7 Mar	**Subject Overview** **Introduction to Operating Systems**	Welcome Module Week 1 Module **TB** 1.1, 1.3.1, 1.3.2, 1.3.4, 1.5.1, 1.5.2, 1.6 (up to 1.6.1)	**Tutorials start** **(even those before the first lecture)**		Sun 9 Mar 11:59pm
2 10 Mar - 14 Mar	**Processes and Threads: Introduction** **Processes and Threads: Interprocess Communication**	Week 2 Module **TB** 2.1, 2.2.1, 2.2.2, 1.6.1 **TB** 2.3.1 to 2.3.4, 2.3.6	**University holiday on Monday, 10 Mar** (tutorials will be rescheduled)		Sun 16 Mar 11:59pm
3 17 Mar - 21 Mar	**Processes and Threads: Interprocess Communication (continued)** **Processes and Threads: Scheduling**	Week 3 Module **TB** 2.3.1-2.3.4, 2.3.6, 2.4.1-2.4.3 (up to guaranteed scheduling)			Tutorial day at 11:59pm

Full subject schedule in Canvas

Textbooks

1. Tanenbaum, A.S. & Herbert Bos (2015). *Modern operating systems (Fourth Edition)*. Pearson/Prentice Hall. ISBN: 9781292061429.
2. Kurose, J.F and Ross, K.W. (2017). *Computer Networking: A Top-Down Approach (Seventh Edition)*. Addison Wesley. ISBN: 9781292153605.

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Assessment

1. **Weekly quizzes – 10%**
   - 12 quizzes, 1 mark each, for a maximum cumulative quiz mark of 10
   - Online with unlimited attempts
2. **Projects – 30%**
   - Programming projects in C
3. **Mid-semester test – 10%**
4. **Exam – 50%**
   - Written, on-campus, closed-book

Week	Quiz	Project	MST	Exam
1	✓			
2	✓			
3	✓			
4	✓	P1 Out		
5	✓			
6	✓		✓	
7	✓	(P1) ✓		
8	✓	P2 Out		
9	✓			
10	✓			
11	✓	(P2) ✓		
12	✓			
--				✓

Projects

- 2 programming projects in C
- Project 1: individual
- Project 2: teams of 2
- Structured around scaffolded tasks, from simple to complex
- Marking is primarily automated – **Must** follow the specs
- We give you some of the tests we use for marking, use them!

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Project Extensions

- 1-3 business days:
  - Have an AAP or fill in FEIT's student declaration form (https://q.surveys.unimelb.edu.au/jfe/form/SV_cOanjMgfzZ2iSqy)
  - **Apply** by submitting COMP30023's **extension request** form (https://forms.office.com/r/hRjkf82uxR)
- More than 4 business days:
  - Apply through the special consideration portal (https://students.unimelb.edu.au/your-course/manage-your-course/exams-assessments-and-results/special-consideration)
  - Submit your **successful outcome** via COMP30023's **extension outcome** form (https://forms.office.com/r/SYNsEGB3kG)

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Mid-Semester Test

- Week 6, Thu 10/04, Kwong Lee Dow Building 122A and 101A
- 2 sessions: 2pm - 3pm and 3pm – 4pm
- Form to sign up for a session will be released tentatively in Week 3. Allocations will be done on a first-come, first-served basis
- Written, on-campus, closed-book
- 50 minutes
- Includes content covered up to the end of Week 5

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MST Extensions

- No extensions granted for the MST
- If you are experiencing circumstances that prevent you from attending your scheduled MST session, please **apply for special consideration for an alternative outcome**
- Once you receive a successful outcome for your special consideration application, please reach out to Maria via email to discuss alternative arrangements

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Weekly Quizzes

- 12 quizzes, each worth 1 mark
- Maximum cumulative quiz mark is 10 – 2 contingency quizzes
- Require an access code given during your tutorial
- Access codes are not transferable between tutorials (except on Weeks 1, 2, 7 – same code for all tutorials)
- ~~Available until 11:59 pm on the day of your tutorial~~ **Available for 24 hours from the start of your tutorial. E.g., if your tutorial starts at 9am on Monday, you will have until 9am on Tuesday to finish the quiz (except on Weeks 1, 2, 7 – due Sun 11:59pm)**
- Available as Canvas quizzes
- Unlimited attempts. Final mark is the highest of all attempts
- Mark is 0 or 1

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Weekly Quiz Extensions

- No extensions for weekly quizzes
- 2 contingency quizzes
- If you are experiencing circumstances that prevent you from taking more than 2 quizzes, you must apply for special consideration

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Hurdle Requirements

Exam component

- 30/60 for the mid-semester test and end-of-semester written examination combined.

Non-exam component

- 20/40 combined over non-exam components.

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Academic Integrity

- Academic Integrity Expectations and Information on Canvas
- Academic Integrity is of utmost importance, and we will all follow the policies of the University and of FEIT
- For the purpose of ensuring academic integrity, all submission attempts by a student may be inspected, regardless of the number of attempts made
- Further information: http://academicintegrity.unimelb.edu.au.

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Communication

Teaching staff → Students

- LMS/Ed announcements
- Lectures

Students → Staff

- Check the LMS for announcements
- Lectures and tutorials
- Ed discussion forum
- Consultation hours
- Personal matters only: send an email *with COMP30023 in the subject line*

Tips for Success

- **Understand** the material, don't just memorise it
- If you fall behind, try to catch up as fast as possible
- Attempt the tutorial and practical tasks every week
- Check the LMS announcements and the discussion board posts
- Talk to other students

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Class Representatives

- 2-3 volunteers to be class representatives
- The role involves:
  - Attending a meeting with colleagues in the School
  - Acting as an informal liaison between students and teaching staff
- Something to add to your resumé
- We would need to publish your name and email address on the subject's Canvas page
- **Send an email to Maria by March 7** with a couple of sentences about you and why you are interested in becoming a student representative

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What's on this Week?

- Tutorials start this week!
- Setup access to your VMs
  - Submit the VM Access assignment on Canvas
  - **Submit by Friday 7/04, 11:59pm** to get the IP address of your VM by Monday (needed for practicals next week!)
  - If you have tutorials later in the week, you can submit by March 9 (Sunday, 11:59pm)

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